

LANSING, MI, USA

WMO: 725390

Lat: 42.776N

Lon: 84.600W

Elev: 859

StdP: 14.25

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14836

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -1.0	(c) 4.3	(d) -7.7	(e) 3.8	(f) 0.5	(g) -2.6	(h) 4.9	(i) 5.7	(j) 26.6	(k) 26.1	(l) 24.6	(m) 26.0	(n) 6.2	(o) 280	(p) 0.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	89.3	72.9	86.4	71.5	83.8	69.9	75.8	85.0	74.1	82.6	72.4	80.4	11.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.9	126.1	81.2	71.2	118.8	79.0	69.6	112.4	77.4	39.8	85.1	38.1	82.8	36.5	80.3	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	20.5	18.5	-8.8	93.7	6.5	3.0	-13.5	95.8	-17.2	97.6	-20.9	99.3	-25.5	101.5
			-9.3	78.8	6.3	2.3	-13.8	80.5	-17.5	81.8	-21.0	83.1	-25.6	84.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.3	23.2	25.4	34.7	46.7	58.0	67.7	71.4	69.7	62.7	50.8	39.2	29.2	
	DBStd	19.21	11.45	10.97	11.31	9.28	8.70	6.98	5.94	5.76	7.75	8.72	9.31	9.73	
	HDD50	3299	831	690	493	169	22	0	0	0	4	98	344	648	
	HDD65	6702	1295	1109	940	551	254	50	13	22	135	448	774	1111	
	CDD50	2695	1	1	20	71	269	533	662	609	384	123	20	2	
	CDD65	623	0	0	1	3	36	132	211	167	65	8	0	0	
	CDH74	5599	0	0	11	45	367	1227	1980	1345	560	64	0	0	
	CDH80	1627	0	0	1	5	84	382	650	353	144	8	0	0	
Wind	WSAvg	8.7	10.2	10.1	9.7	10.0	8.7	7.5	6.9	6.5	6.9	8.5	9.5	9.7	
Precipitation	PrecAvg	30.6	1.5	1.4	2.2	2.9	3.0	3.5	2.6	3.1	3.3	2.2	2.6	2.1	
	PrecMax	39.6	3.3	3.8	4.4	5.8	10.4	10.2	6.4	9.8	8.3	5.7	5.6	4.3	
	PrecMin	21.2	0.2	0.2	0.2	0.7	0.6	0.2	0.5	0.2	0.0	0.4	0.5	0.4	
	PrecStd	4.3	0.9	0.9	1.0	1.2	1.9	1.9	1.2	1.6	1.9	1.2	1.2	1.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	58.2	74.1	79.6	86.4	91.4	93.4	91.3	88.7	80.4	68.1	59.0	
		MCWB	52.8	52.2	61.9	61.5	69.0	74.2	75.5	73.9	71.5	67.7	57.3	55.4	
	2%	DB	49.0	51.4	65.1	73.3	82.3	87.8	89.8	86.9	84.2	74.9	61.9	52.0	
		MCWB	45.8	46.1	56.4	58.7	66.6	71.4	74.0	72.5	69.6	63.0	54.5	48.8	
	5%	DB	42.8	44.7	58.0	68.3	78.3	84.7	86.5	83.8	80.3	70.3	58.1	46.9	
		MCWB	39.9	40.2	50.8	55.3	64.6	69.8	72.1	70.6	67.4	61.3	51.9	43.5	
	10%	DB	38.0	39.8	52.0	63.5	73.9	81.1	83.5	81.1	76.7	66.1	53.9	41.8	
		MCWB	35.8	36.4	45.4	53.4	62.4	67.9	70.1	69.2	65.6	58.3	48.4	38.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.3	54.1	62.3	66.1	72.1	76.3	79.3	77.5	75.0	70.2	60.1	56.1
			MCDB	55.1	56.5	73.2	76.5	82.7	87.6	88.8	87.3	84.8	78.3	65.2	59.2
2%		WB	46.4	46.6	57.1	61.0	69.1	73.8	76.7	75.0	72.2	66.1	56.5	49.7	
		MCDB	48.2	50.5	63.6	68.8	78.0	83.9	85.9	82.5	79.2	71.9	60.4	52.2	
5%		WB	40.1	40.8	52.1	57.5	66.8	71.9	74.6	73.1	69.9	62.4	52.5	43.5	
		MCDB	42.4	44.3	57.6	66.4	75.2	81.3	83.3	80.4	76.7	68.5	57.3	46.2	
10%		WB	35.8	36.6	45.9	53.9	64.1	70.1	72.6	71.3	67.7	59.2	48.4	38.7	
		MCDB	38.0	39.7	51.5	62.3	72.0	78.2	80.6	78.4	74.9	65.3	53.3	41.4	
Mean Daily Temperature Range	5% DB	MDBR	13.2	14.7	17.6	20.8	20.7	20.4	21.0	19.9	21.0	18.4	14.6	11.8	
		MCDBR	16.3	18.6	24.8	27.0	24.2	23.3	22.7	21.6	24.1	24.1	19.9	17.0	
		MCWBR	14.4	15.3	17.2	15.9	12.8	10.8	10.7	9.9	12.1	14.5	14.5	14.6	
	5% WB	MCDBR	15.6	17.4	23.0	23.7	21.0	19.8	20.0	18.1	19.9	21.0	18.2	16.1	
		MCWBR	14.4	15.2	17.1	16.0	12.8	10.7	11.0	9.7	11.6	14.6	15.0	14.7	
Clear-Sky Solar Irradiance	taub	0.295	0.350	0.337	0.384	0.414	0.436	0.425	0.419	0.381	0.344	0.314	0.294		
	taud	2.359	2.163	2.370	2.261	2.227	2.202	2.239	2.250	2.345	2.445	2.476	2.414		
	Ebn at Noon	267	263	287	280	273	266	268	266	269	266	259	257		
	Edh at Noon	26	37	34	41	44	45	43	41	35	28	23	22		
All-Sky Solar Radiation	RadAvg	454	736	1138	1450	1694	1912	1941	1661	1337	851	525	362		
	RadStd	41	93	117	132	157	119	92	79	125	85	69	36		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page